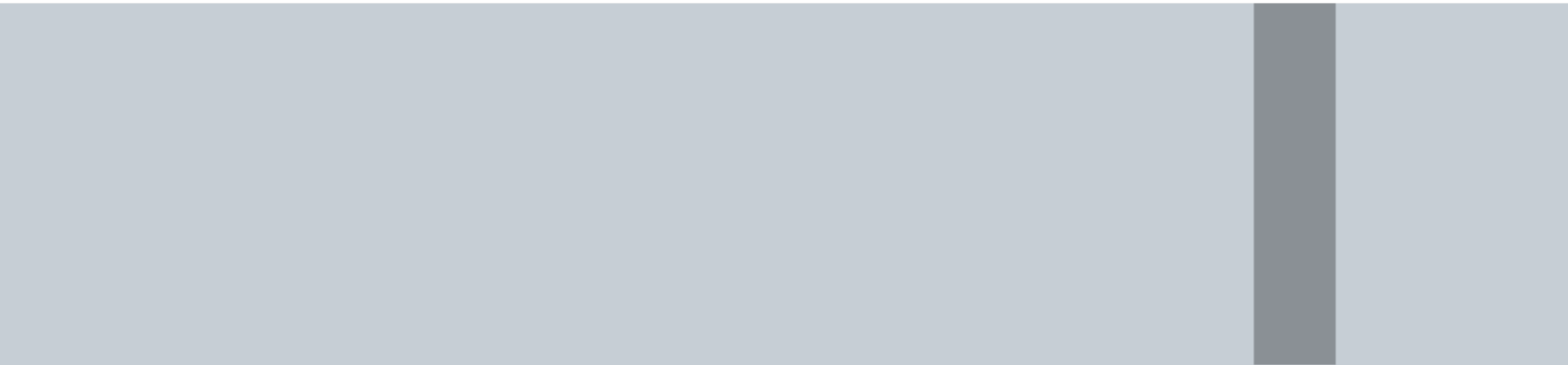
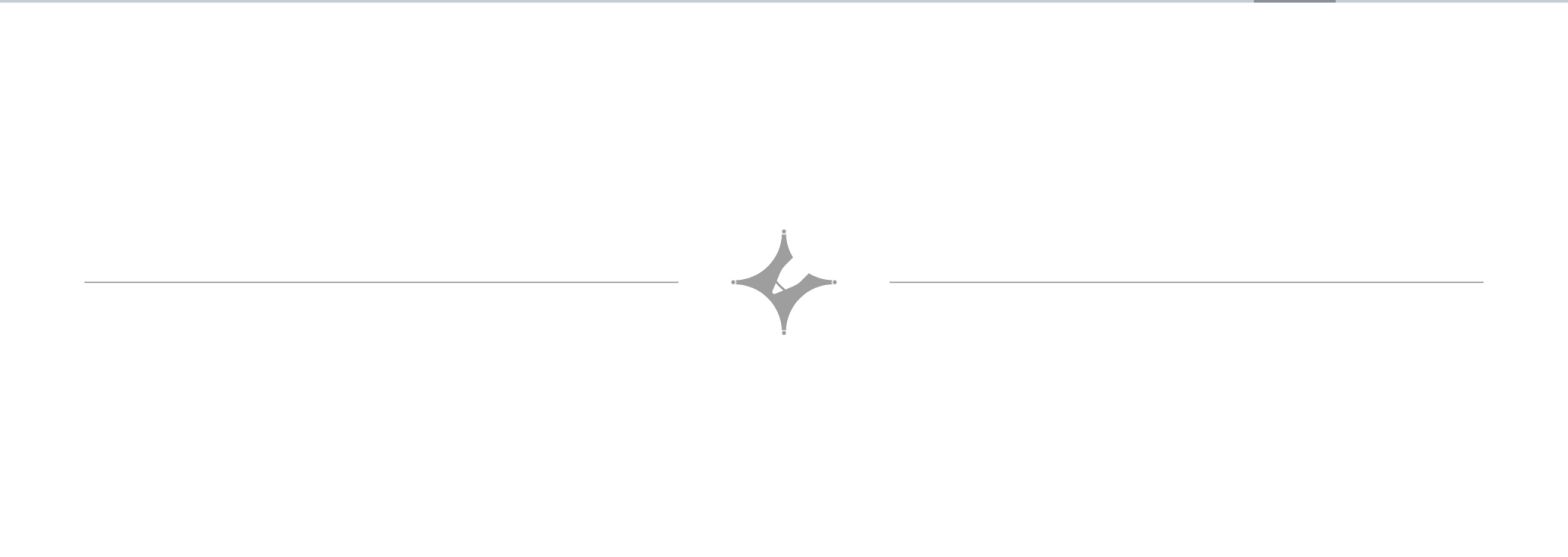
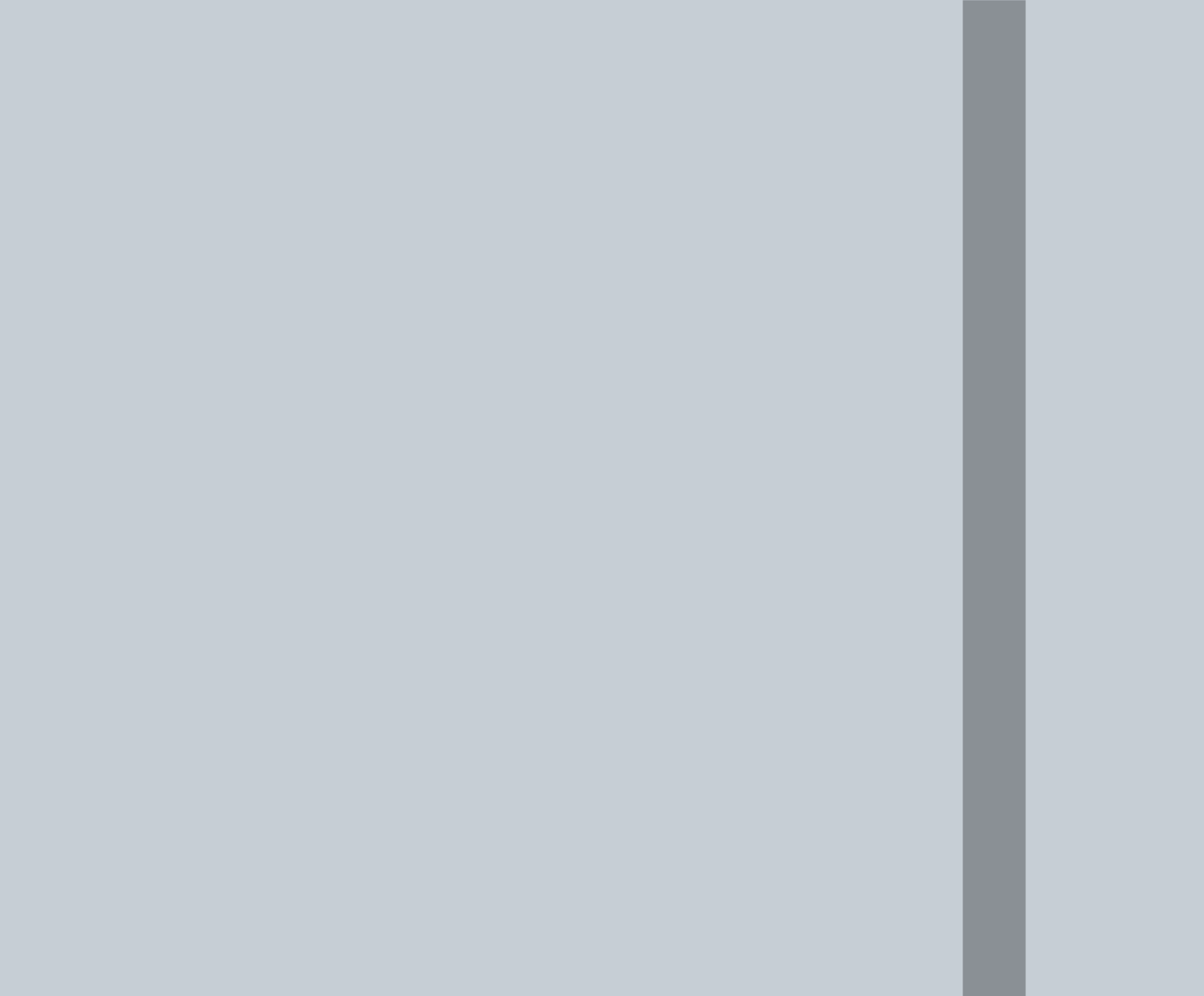
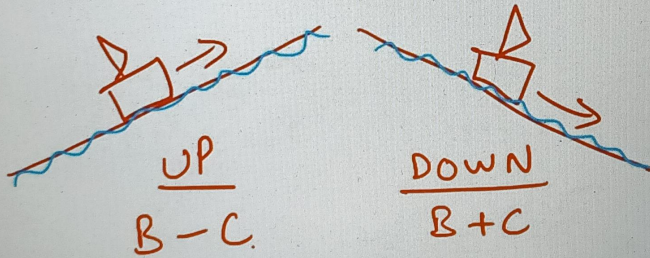




Boats and Streams

CONCEPT

sp. of stream = C .



and rest of the formulas are the same -
(of Speed, dist. and time).

2. A boat rows 16 km up the stream and 30 km downstream taking 5 hours each time. The speed of the current is

- A) 1.1 kmph B) 1.2 kmph C) 1.4 kmph D) 1.5 kmph

$\hookrightarrow d = S \times t$. (B = sp. of boat, C = sp. of current)

$$\Rightarrow C = \frac{d}{t} \Rightarrow (B - C) = \frac{16}{5} \rightarrow \text{① (upstream)}$$

$$\Rightarrow (B + C) = \frac{30}{5} \rightarrow \text{downstream} \rightarrow \text{②}$$

By ① & ② $\Rightarrow B = 1.4 \text{ km/hr}$

3. A motorboat went downstream for 28km and immediately returned. It took the boat twice as long to make the return trip. If the speed of the river flow were twice as high, the trip downstream and back would take 672 min. Find the speed of the boat in still water and the speed of the river flow.

- A) 9 kmph, 3 kmph B) 9 kmph, 6 kmph
C) 8 kmph, 2 kmph D) 12 kmph, 3 kmph

options se pata chal gaya no need to solve further.

$$\Rightarrow T_U = 2 T_D$$

$$\Rightarrow \frac{28}{(B - U)} = \frac{2 \times 28}{(B + U)}$$

$$\Rightarrow B + U = 2B - 2U$$

$$\Rightarrow \underline{3U = B}$$

4. A boat goes downstream at x km/h and upstream at y km/h. The sp. of the boat in still water is

- A) $0.5(x+y)$ B) $0.5(x-y)$ C) $x+y$ D) $x-y$

$$\begin{aligned} B + C &= x \\ B - C &= y \\ \oplus \\ 2B &= x + y \\ B &= \frac{x+y}{2} \end{aligned}$$

5. A boat running upstream takes 8 hours to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of the water current respectively?

- A) 1 : 3 B) 2 : 1 C) 1 : 2 D) 3 : 1

$$\begin{aligned} T &= 8, d = d \\ S &= B - C \end{aligned}$$

$$2(B - C) = \frac{d}{4}$$

($\therefore B - C = \frac{d}{8}$)

$$\Rightarrow 2B - 2C = B + C$$

$$\begin{aligned} T &= 4, d = d \\ S &= B + C \\ B + C &= \frac{d}{4} \end{aligned}$$

$\Rightarrow B = 3C$
 $\hookrightarrow \text{Ratio} \neq 1:3 \text{ but}$
 $\Rightarrow \frac{B}{C} = \frac{3}{1} \text{ So, it's } 3:1$

6. A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is:

- A) 2 mph B) 2.5 mph C) 3 mph D) 4 mph

$$T_U - T_D = \frac{90}{60} = 1.5$$

$$\frac{36}{B-C} - \frac{36}{B+C} = \frac{3}{2}$$

$$\frac{1}{10-C} - \frac{1}{10+C} = \frac{3}{2 \times 36} = \frac{1}{24}$$

$$\frac{10+C - 10+C}{10^2 - C^2} = \frac{1}{24}$$

$$2C \times 24 = 100 - C^2$$

$$C^2 + 48C - 100 = 0$$

$$C^2 + 50C - 2C - 100 = 0$$

$$C(C+50) - 2(C+50) = 0$$

$$(C+50)(C-2) = 0$$

$$C = -50 \text{ or } \underline{\underline{+2}}$$

